

Network Modeling (NM)

Introduction to the concept of “Modeling”

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Network modeling



WHAT



WHY



HOW





Challenge!

Objective:

Propose a model to represent and study the **network of acquaintances in this classroom**

Don't: *draw a map yet, nor list names*

Do: define how you will *represent the system* and what you *intend to learn from it*

Your Proposal Must Define the Following:

- 1. The Performance Metrics: what are the aspects of interest of your network?**
2. The Components: what are the nodes?
3. The Interactions: when are nodes connected?
4. The Methodology: how do you collect data to define the model?

Network modeling



WHAT



WHY



HOW

WHAT

Theoretical or conceptual *framework*

that makes it possible to

represent and study

various aspects of networked systems, including their

components, interactions, and performance

Network modeling



WHAT

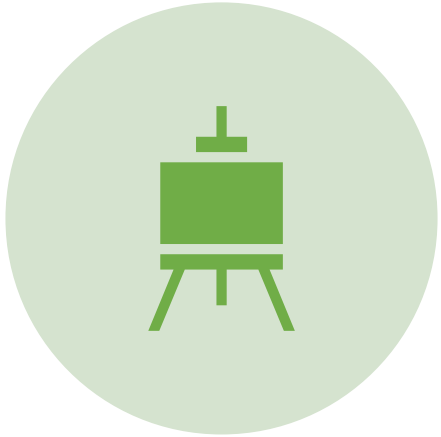


WHY



HOW

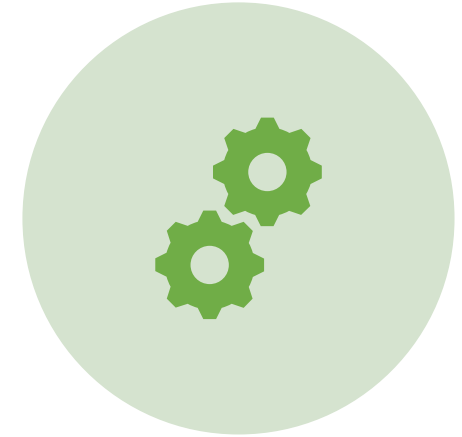
(network) Models are essential for...



DESIGNING



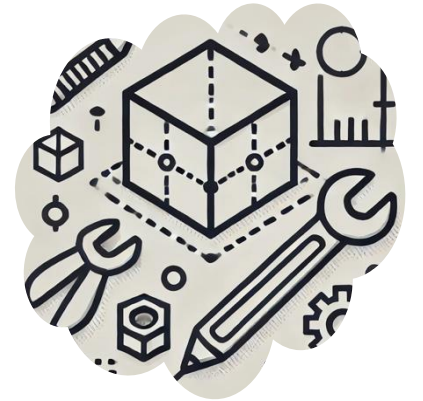
EVALUATING



OPTIMIZING

Advantages of (network) modeling

- Simpler, quicker, and cheaper than prototyping
- Easier to control and single out the effect of each element
 - Gain better understanding of complex protocol inter-relations
- Allow for rough system dimensioning
 - Shall I look for 1Mbit/s or 1Gbit/s solution?
 - How would performance scale with size?



Limits of (network) models

Models are often hyper-simplified

- Easiness is paid back in terms of accuracy
- Simplified model → neglects details that may turn out to be critical in real world → potentially unrealistic performance predictions

Scalability

- Models may not scale effectively for large or highly complex networks
- Computational resources and time may become prohibitive as network size increases

Evolving Technologies

- Rapid changes in network technologies and protocols can render models outdated quickly
- Models may struggle to keep up with the pace of innovation in networking